

M-finescan: vanilla vs orthogonal AL-RNN

Sweep: M in {10, 11, 14, 15} x P in {2, 3, 4, 5, 6}

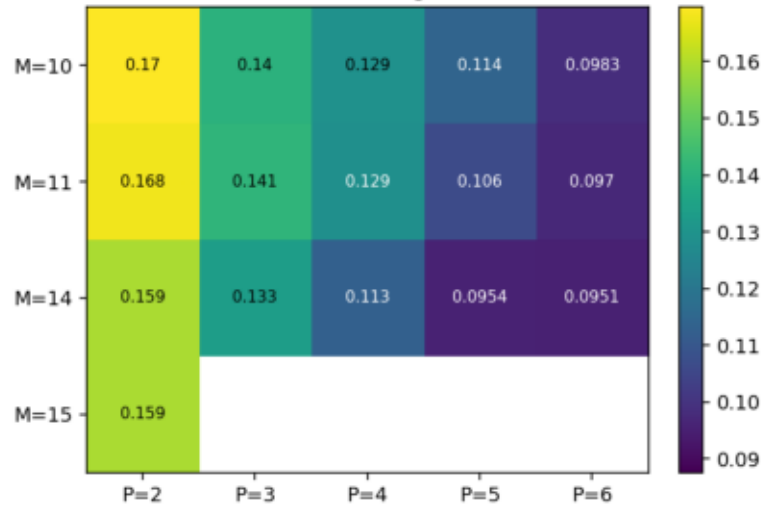
Epochs: 1000 alpha: 1.0 shared frozen B per M seed: 42

Metrics:

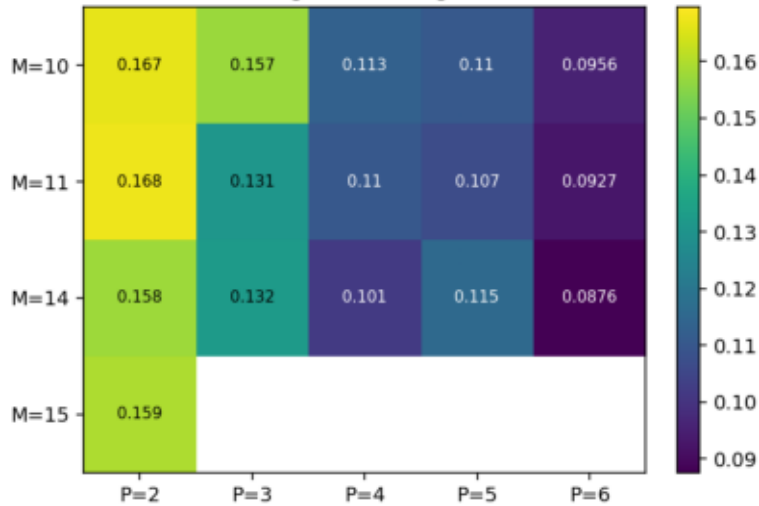
- rmse_train_final - sqrt(final training MSE), one-step prediction
- rmse_h100 - free-run RMSE on the first 100 steps after $t=0$
- Dstsp - $KL(p_{\text{true}} || p_{\text{gen}})$ on a 30^3 binned 3-D density
- PSE - mean Hellinger distance between smoothed power spectra
(the AL-RNN paper's attractor-quality metrics)

training MSE (lower is better)

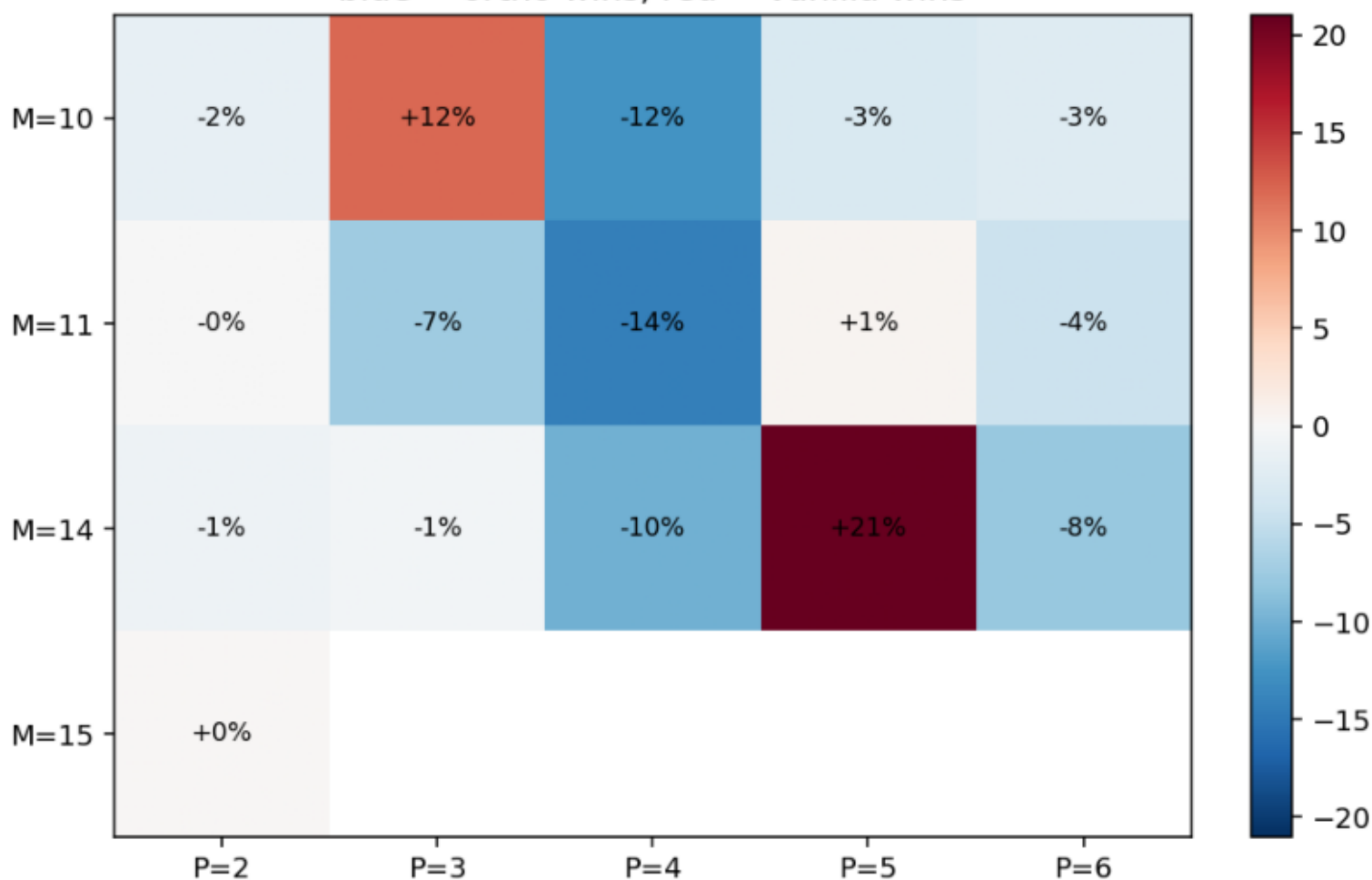
Vanilla: training MSE



Orthogonal: training MSE

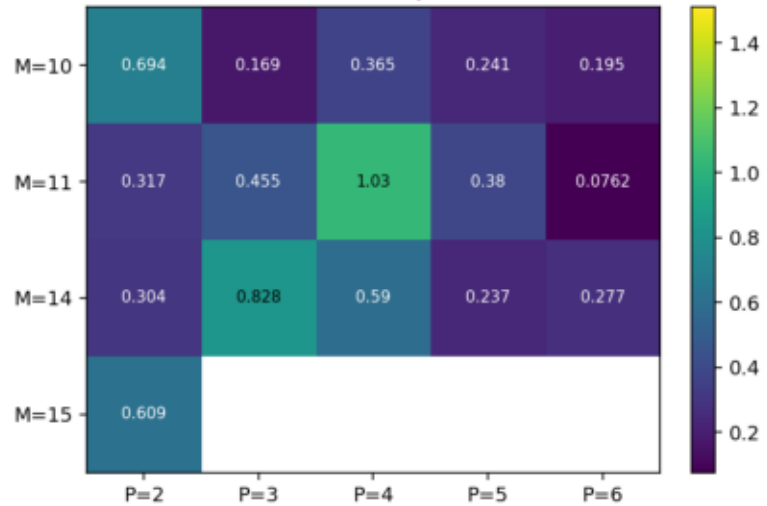


Gap (ortho - vanilla) on training MSE
blue = ortho wins, red = vanilla wins

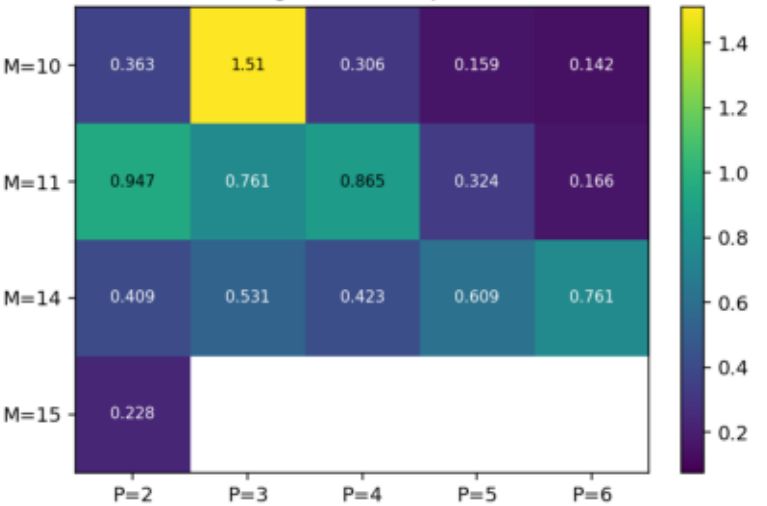


100-step RMSE (lower is better)

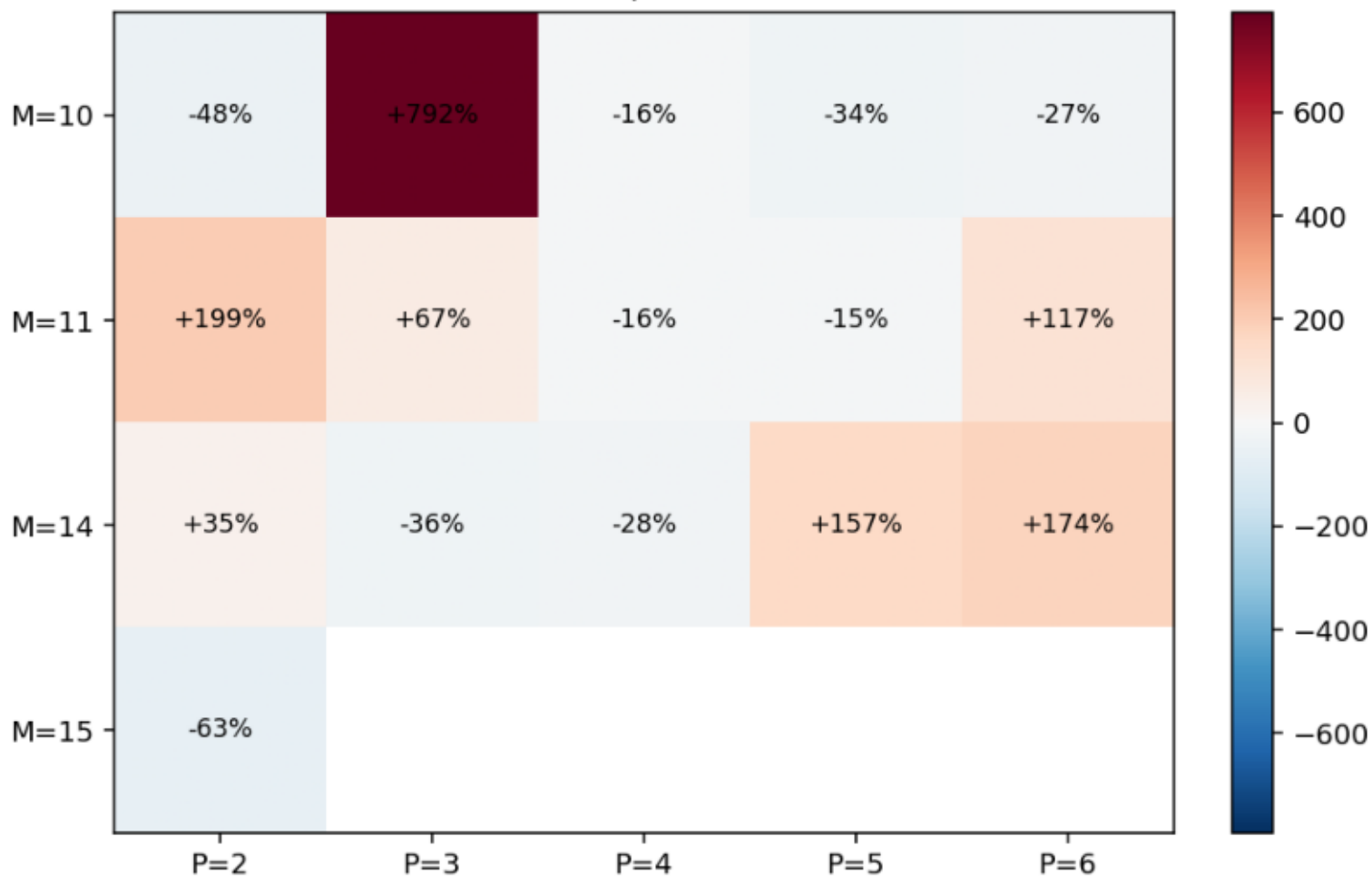
Vanilla: 100-step RMSE



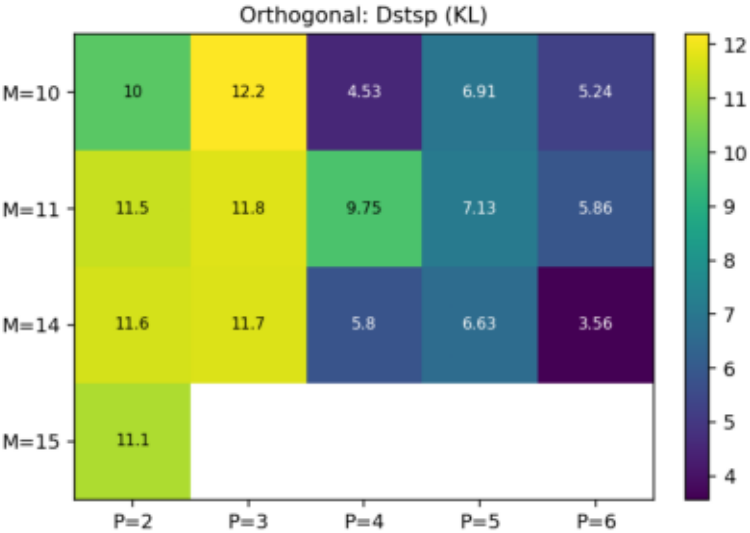
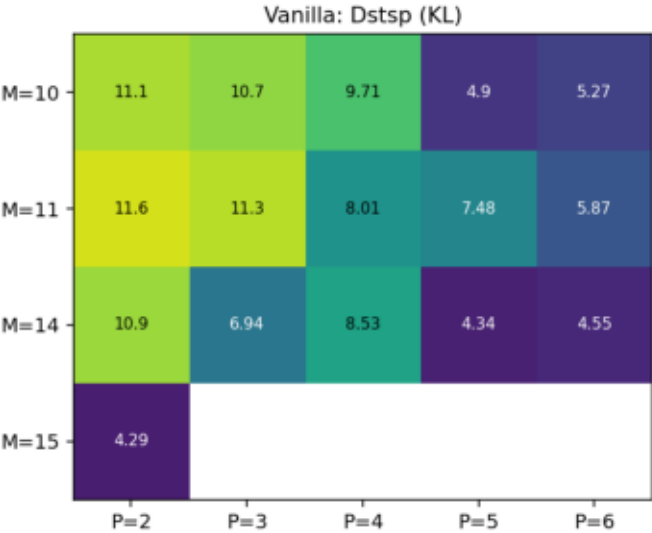
Orthogonal: 100-step RMSE



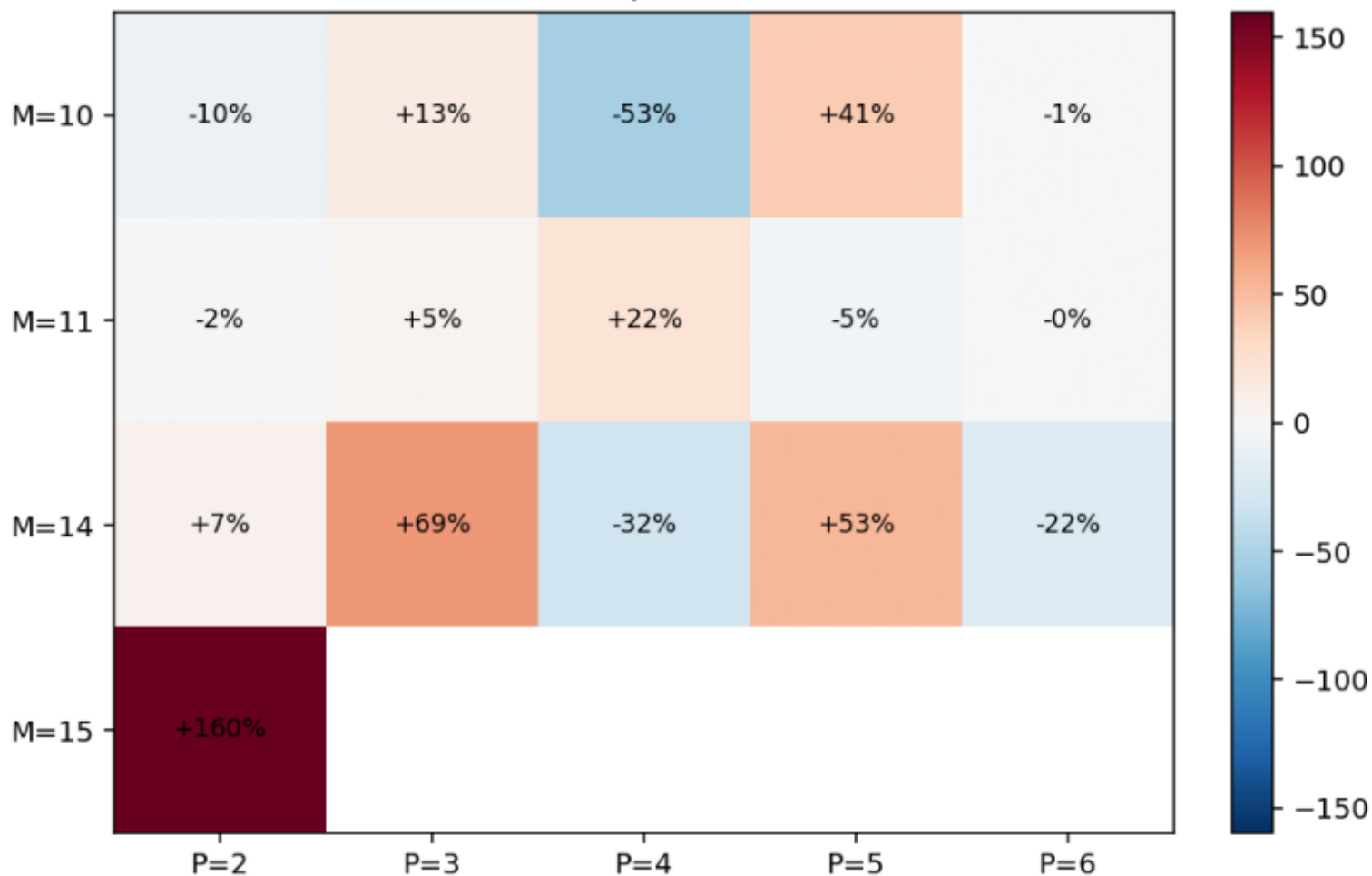
Gap (ortho - vanilla) on 100-step RMSE
blue = ortho wins, red = vanilla wins



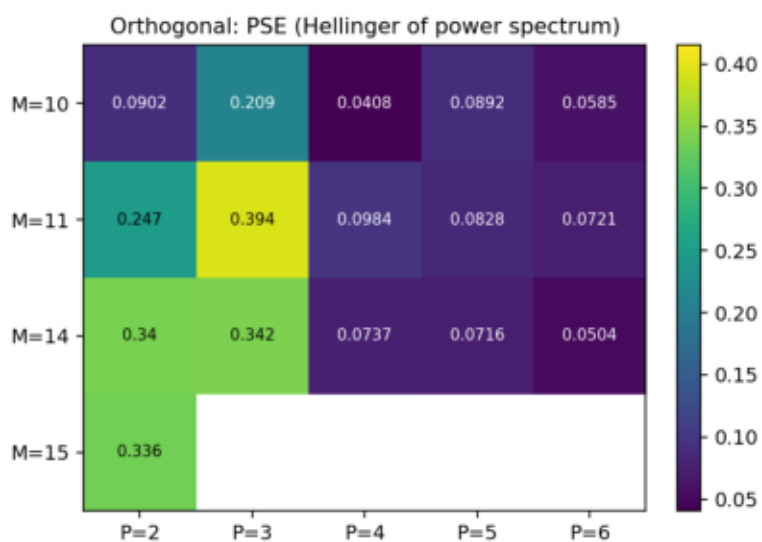
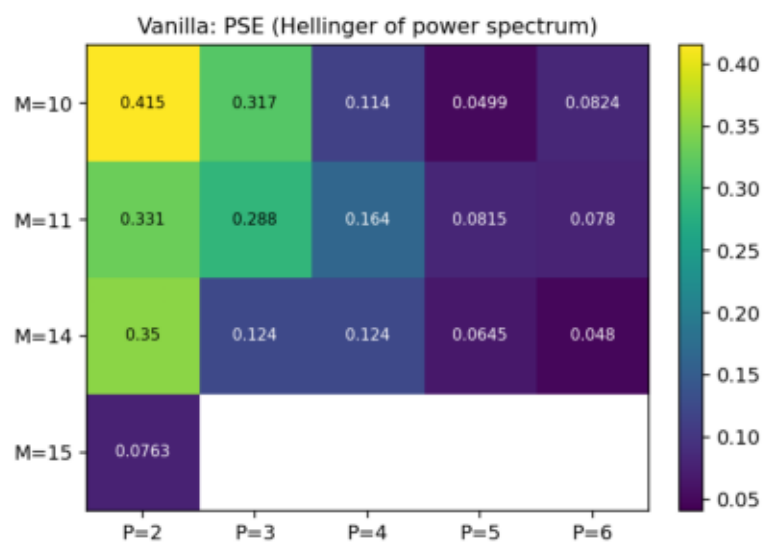
Dstsp (KL) (lower is better)



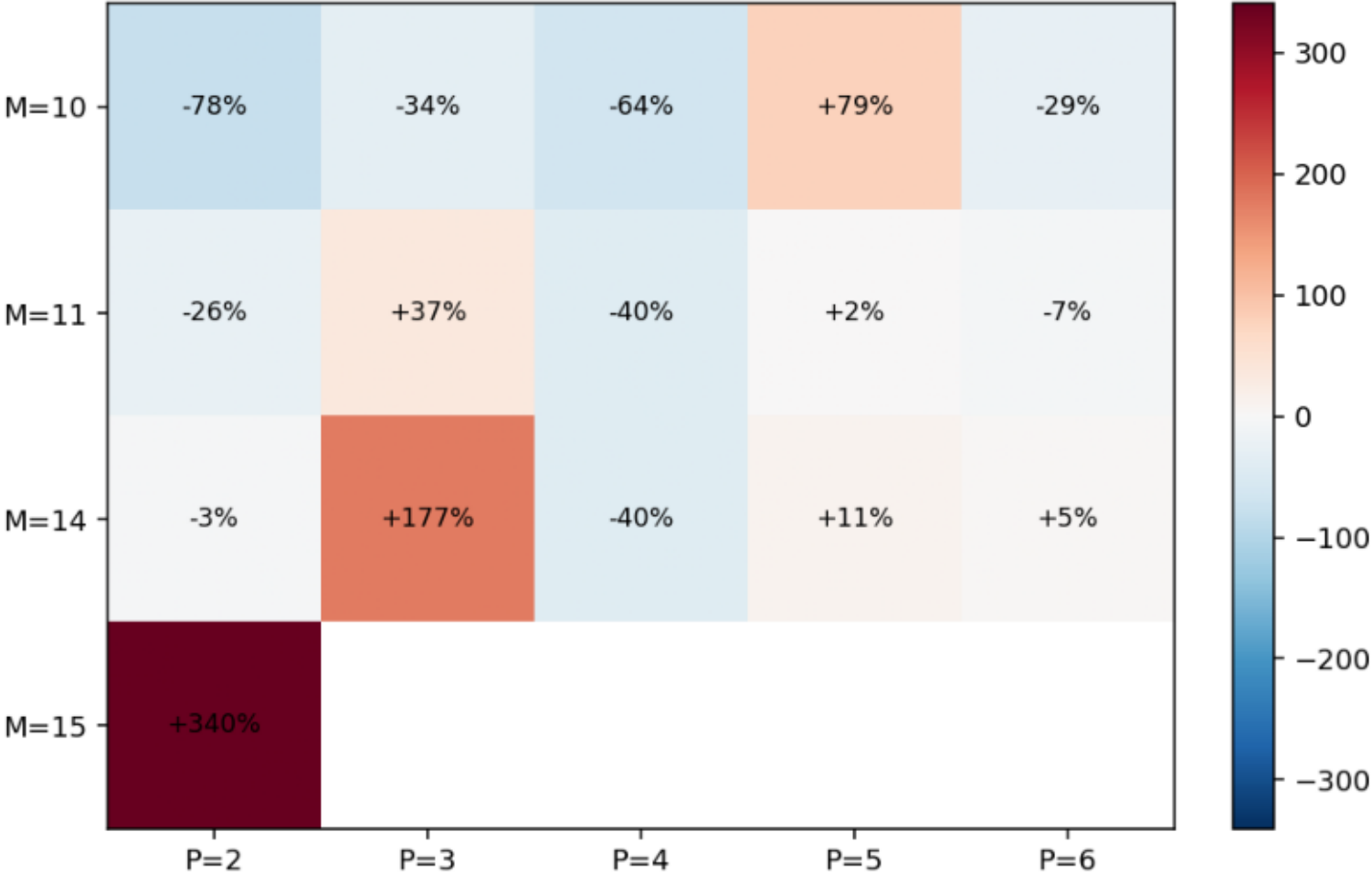
Gap (ortho - vanilla) on Dstsp (KL)
blue = ortho wins, red = vanilla wins



PSE (Hellinger of power spectrum) (lower is better)



Gap (ortho - vanilla) on PSE (Hellinger of power spectrum)
blue = ortho wins, red = vanilla wins



Metrics vs M, by P (vanilla dashed, ortho solid)

